

RWA Hamiltonian Working

First we begin with the original Hamiltonian:

$$\begin{aligned} H_{NV} = & (D_0 + \epsilon_{||}\sigma_{||})S_z^2 + \gamma_{NV}B_{||}S_z \\ & + \gamma_{NV}B_{\perp}S_x + \epsilon_{\perp}\sigma_x(S_x^2 - S_y^2) \\ & + \epsilon_{\perp}\sigma_y(S_xS_y + S_yS_x) \end{aligned}$$

Let's consider a microwave drive term:

$$H_{\text{drive}} = 2\Omega \cos(\omega_d t) S_x$$

Where:

Ω = drive amplitude

ω_d = drive frequency

Now let's consider the Unitary rotating frame:

$$U(t) = e^{-i\omega_d S_z t}$$

So we are rotating around S_z at the drive frequency ω_d .

Assumptions:

- Dropped fast oscillating terms
- Keep only near-resonant ($\Delta m = \pm 1$) terms that become time-independent in the rotating frame
- Terms that change m by ± 2 (e.g. $(S_x^2 - S_y^2)$) rotate at $\pm 2\omega_d$ in this frame

Define the rotating-frame Hamiltonian:

$$H_{\text{rot}}(t) = U^\dagger(t) [H_0 + H_{\text{drive}}(t)] U(t) - iU^\dagger(t)\dot{U}(t)$$

where H_0 is the static Hamiltonian (original expression)

Let's consider the spin matrices under these transformations. Using Baker-Campbell-Hausdorff series for:

$$A = -i\omega_d t S_z$$

We get:

$$e^A S_x e^{-A} = S_x + [A, S_x] + \frac{1}{2!}[A, [A, S_x]] + \dots$$

With

$$[S_z, S_x] = iS_y$$

$$[S_z, [S_z S_x]] = -[S_z, iS_y]$$

$$= -S_x$$

Hence:

$$U^\dagger S_z U = S_z$$

$$U^\dagger S_x U = \cos(\omega_d t) S_x - \sin(\omega_d t) S_y$$

$$U^\dagger S_y U = \sin(\omega_d t) S_x + \cos(\omega_d t) S_y$$

$$U^\dagger (S_x^2 - S_y^2) U = \cos(2\omega_d t) (S_x^2 - S_y^2) - \sin(2\omega_d t) (S_x S_y + S_y S_x)$$

$$U^\dagger (S_x S_y + S_y S_x) U = \sin(2\omega_d t) (S_x^2 - S_y^2) + \cos(2\omega_d t) (S_x S_y + S_y S_x)$$

Finally:

$$-i\dot{U}^\dagger U = -iU^\dagger (-i\omega_d S_z U)$$

$$= -\omega_d S_z$$

Since $U^\dagger S_z U = S_z$

Now we can build $H_{rot}(t)$ term by term.

Diagonal Term:

$$\begin{aligned} & (D_0 + \epsilon_{||} \sigma_{||}) S_z^2 + \gamma_{NV} \beta_{||} S_z \\ & \Rightarrow \\ & (D_0 + \epsilon_{||} \sigma_{||}) S_z^2 + \gamma_{NV} \beta_{||} S_z - \omega_d S_z \end{aligned}$$

Static + Transverse Field Term:

$$\begin{aligned} & \gamma_{NV} \beta_{\perp} S_x \\ & \Rightarrow \\ & \gamma_{NV} \beta_{\perp} (\cos \omega_d t S_x - \sin \omega_d t S_y) \end{aligned}$$

Stress Terms:

1.

$$\begin{aligned} & \epsilon_{\perp} \sigma_x (S_x^2 - S_y^2) \\ & \Rightarrow \\ & \epsilon_{\perp} \sigma_x [\cos(2\omega_d t)(S_x^2 - S_y^2) - \sin(2\omega_d t)(S_x S_y + S_y S_x)] \end{aligned}$$

2.

$$\begin{aligned} & \epsilon_{\perp} \sigma_y (S_x S_y + S_y S_x) \\ & \Rightarrow \\ & \epsilon_{\perp} \sigma_y [\sin(2\omega_d t)(S_x^2 - S_y^2) + \cos(2\omega_d t)(S_x S_y + S_y S_x)] \end{aligned}$$

Drive Term:

$$\begin{aligned} U^{\top} (2\Omega \cos(\omega_d t) S_x) U &= 2\Omega \cos(\omega_d t) (\cos \omega_d t S_x - \sin \omega_d t S_y) \\ &= \Omega ((1 + \cos(2\omega_d t)) S_x - \sin(2\omega_d t) S_y) \end{aligned}$$

Putting it all together:

$$\begin{aligned} H_{\text{rot}}(t) &= (D_0 + \epsilon_{\parallel} \sigma_{\parallel}) S_z^2 + (\gamma_{\text{NV}} B_{\parallel} - \omega_d) S_z \\ &+ [\gamma_{\text{NV}} B_{\perp} \cos \omega_d t + \Omega(1 + \cos 2\omega_d t)] S_x - [\gamma_{\text{NV}} B_{\perp} \sin \omega_d t + \Omega \sin 2\omega_d t] S_y \\ &+ \epsilon_{\perp} [\sigma_x (\cos 2\omega_d t (S_x^2 - S_y^2) - \sin 2\omega_d t (S_x S_y + S_y S_x)) + \sigma_y (\sin 2\omega_d t (S_x^2 - S_y^2) + \cos 2\omega_d t (S_x S_y + S_y S_x))] \end{aligned}$$

Converting to 3×3 Matrix:

$$H_{\text{rot}}(t) = \begin{pmatrix} H_{00}(t) & H_{01}(t) & H_{02}(t) \\ H_{10}(t) & H_{11}(t) & H_{12}(t) \\ H_{20}(t) & H_{21}(t) & H_{22}(t) \end{pmatrix}$$

Diagonal entries:

$$H_{00}(t) = D_0 + \epsilon_{\parallel} \sigma_{\parallel} + \gamma_{\text{NV}} B_{\parallel} - \omega$$

$$H_{11}(t) = 0$$

$$H_{22}(t) = D_0 + \epsilon_{\parallel} \sigma_{\parallel} - \gamma_{\text{NV}} B_{\parallel} + \omega$$

Off-diagonal: Let:

$$C(t) \equiv \frac{\gamma_{\text{NV}} B_{\perp}}{2} + \Omega \cos(\omega t)$$

Then:

$$H_{01}(t) = \sqrt{2} C(t) e^{+i\omega t}, \quad H_{10}(t) = \sqrt{2} C(t) e^{-i\omega t}$$

$$H_{12}(t) = \sqrt{2} C(t) e^{+i\omega t}, \quad H_{21}(t) = \sqrt{2} C(t) e^{-i\omega t}$$

Matrix elements that couple $|+1\rangle \leftrightarrow |-1\rangle$ ($\Delta m = \pm 2$):

$$H_{02}(t) = \epsilon_{\perp} (\sigma_x \cos \phi + \sigma_y \sin \phi + i(-\sigma_x \sin \phi - \sigma_y \cos \phi))$$

$$H_{20}(t) = \epsilon_{\perp} (\sigma_x \cos \phi + \sigma_y \sin \phi + i(-\sigma_x \sin \phi + \sigma_y \cos \phi))$$

Or equivalently:

$$H_{02}/H_{20} = \frac{\epsilon_{\perp}}{2} \left[(\sigma_x + i\sigma_y) e^{+2i\omega t} + (\sigma_x - i\sigma_y) e^{-2i\omega t} \right]$$

Full Matrix

$$H_{\text{rot}}(t) =$$

$$\begin{pmatrix} D_0 + \epsilon_{\parallel} \sigma_{\parallel} + \gamma_{\text{NV}} B_{\parallel} - \omega & \sqrt{2} C(t) e^{+i\omega t} & \epsilon_{\perp} (\sigma_x \cos(2\omega t) + \sigma_y \sin(2\omega t) + i(-\sigma_x \sin(2\omega t) - \sigma_y \cos(2\omega t))) \\ \sqrt{2} C(t) e^{-i\omega t} & 0 & \sqrt{2} C(t) e^{+i\omega t} \\ \epsilon_{\perp} (\sigma_x \cos(2\omega t) + \sigma_y \sin(2\omega t) + i(-\sigma_x \sin(2\omega t) + \sigma_y \cos(2\omega t))) & \sqrt{2} C(t) e^{-i\omega t} & D_0 + \epsilon_{\parallel} \sigma_{\parallel} - \gamma_{\text{NV}} B_{\parallel} + \omega \end{pmatrix}$$